Ben Lonnqvist

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Education

2020- Ph.D. Candidate, Neuroscience
EPFL (Swiss Federal Institute of Technology in Lausanne) | Lausanne, Switzerland
- Laboratory of Psychophysics–PI: Prof Michael Herzog
- Primary interests: neural network modelling of vision; computational neuroscience, psychophysics
- Teaching: Teaching assistant for Real Analysis 2020, 2021
- EPFL candidate for the Google Ph.D. Fellowship in 2020

2016-2020 M.A. Hons, Economics and Finance
University of Aberdeen | Aberdeen, Scotland
- First-class Honours degree; thesis: Optimal search in multi-armed bandit problems
- Quantitative focus in electives
- Relevant coursework: Econometrics, Mathematical and Statistical Methods in Economics, Stochastic processes, Understanding Statistics, Proof-based Microeconomics

Research

Publications

2021 **Lonnqvist, B.**, Bornet, A., Doerig, A., Herzog, M. H. (2021). A comparative biology approach to DNN modeling of vision: A focus on differences, not similarities. Publisher: *Journal of Vision*. DOI: https://doi.org/10.1167/jov.21.10.17

2020 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Crowding in humans is unlike that in convolutional neural networks. Publisher: *Neural Networks*. DOI: https://doi.org/10.1016/j.neunet.2020.03.021

Pre-prints

2022 **Lonnqvist, B.**, Machiraju, H., Herzog, M. H. (2022). A comment on Guo et al. (2022). arXiv preprint. DOI: https://doi.org/10.48550/arXiv.2208.01456.

2020 **Lonnqvist, B.**, Elsner, M., Hunt, A. R., Clarke, A. D. F. (2020). Modeling individual variation in visual search with reinforcement learning. PsyArXiv. DOI: https://doi.org/10.31234/osf.io/suj28.

Conference oral presentations

2022 **Lonnqvist, B.**, Bornet, A., Doerig, A., Herzog, M. H. (2022). Global Information Processing in Feedforward Deep Networks. Oral presentation at the *22nd Annual Meeting of the Vision Sciences Society (VSS)*.

2019 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Object Recognition in Deep Convolutional Neural Networks is Fundamentally Different to That in Humans. Oral presentation at the *18th annual meeting of the Scottish Vision Group (SVG)*.

Conference poster presentations

2021 **Lonnqvist, B.**, Doerig, A., Bornet, A., Francis, G., Schmittwilken, L., Herzog, M. H. (2021). How crowding challenges (feedforward) convolutional neural networks. Poster presented at the *21st Annual Meeting of the Vision Sciences Society (VSS)*. DOI: https://doi.org/10.1167/jov.21.9.2039.

Invited talks

2023 **Lonnqvist, B.**, Herzog, M. H. (2023). Deep Neural Network Models of Figural Perception. Talk presented at the Integrative Neuroscience and Cognition Center (INCC), Université Paris Cité, and the Centre Nationale de la Recherche Scientifique (CNRS) Vision Group meeting.

2019 **Lonnqvist, B.**, Clarke, A. D. F., Hunt, A. (2019). Changes in Human Eye Movement Strategies Over Time. Talk presented at a meeting of the schools of Psychology and Computer Science at the University of Essex, Britain, UK.

Research Work Experience

Aug - 2020 - **EPFL (Swiss Federal Institute of Technology in Lausanne) | Doctoral Assistant**
- Doctoral Research Assistant in the Laboratory of Psychophysics
- Focus on deep learning and psychophysics

Mar - May 2018 - *2020* **Aberdeen School of Psychology | Undergraduate Research Assistant**
- Voluntary research assistant in the Consciousness, Attention, and Perception Lab
- First-authored a paper that has now been published in *Neural Networks*
- Using Python, Keras and NumPy, programmed several architectures of Deep Neural Networks and wrote testing environments to produce novel research on visual crowding in convolutional NNs
- Used SSH tunneling to access a Linux high-power computing cluster

May - Aug 2019 - *2019* **Aberdeen School of Psychology | Research Intern**
- Summer research intern at the Eye Movements and Attention Lab
- Using Python, modeled Bayes-optimal eye movements parameterized by human visual search data
- Using MATLAB and PsychToolbox, wrote a visual search psychophysical experiment
- Ran 65 participant-hours of eye tracking experiments using EyeLink hardware

Awards

Rank Prize Foundation Optoelectronics Studentship

2019 A £3000 studentship awarded for a summer research project.

Research visit support by the James S McDonnell Foundation Scholar Award in Human Cognition (PI: Dr A. Hunt)

2019 A small undergraduate research visit support for a research visit to the University of Essex in July 2019.

Teaching

MSc Thesis | MSc Student Clémentine Lévy-Fidel

2023- Currently advising MSc student Clémentine Lévy-Fidel's MSc thesis on modeling peripheral vision using autoencoder models.

Semester project | MSc Student Merkourios Simos

2022-2023 Supervision of MSc student Merkourios Simos at EPFL on their Computational Neuroscience semester project on doing large-scale psychophysical measurements of deep learning models on the MIT Brain-Score platform.

Computational Neuroscience Minor Thesis | MSc Student Zhengqing Wu

2022-2023 Supervision of MSc student Zhengqing Wu at EPFL on their Computational Neuroscience Minor Project on unsupervised segmentation in variational autoencoders.

Teaching assistant | Real Analysis

2020, 2021 Teaching assistant for the Real Analysis course for undergraduates at EPFL in 2020 and 2021.

Academic service

Reviewing

Journals *Neural Networks, Neural Computation*